5.2



We saw in Section 5.1 that a limit of the form

$$\lim_{n\to\infty}\sum_{i=1}^{n} f(x_i^*)\Delta x$$

$$= \lim_{n\to\infty}[\, f(x_1^*)\Delta x + f(x_2^*)\Delta x + \ldots + f(x_n^*)\Delta x]$$

arises when we compute an area.

In this section, we will learn about:

Integrals with limits that represent a definite quantity.

DEFINITE INTEGRAL

If f is a function defined for $a \le x \le b$,

Then, the definite integral of f from a to b is

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\Delta x$$

Where $\Delta x = (b - a)/n$, provided that this limit exists.

If it does exist, we say f is integrable on $[a, b]$.

The sum above is called a Riemann sum.

NOTATION $\int_a^b f(x)\,dx$

- The symbol $\int$ is called an integral sign.
- $f(x)$ is called the integrand.
- a and b are called the limits of integration;

  a is the lower limit and b is the upper limit.
  - For now, the symbol dx has no meaning by itself; $\int_a^b f(x)\,dx$ is all one symbol.

  The dx simply indicates that the independent variable is x.
  - The procedure of calculating an integral is called integration.

Remark

- The definite integral $\int_a^b f(x)dx$ is a number.

It does not depend on *x.*

where a < b

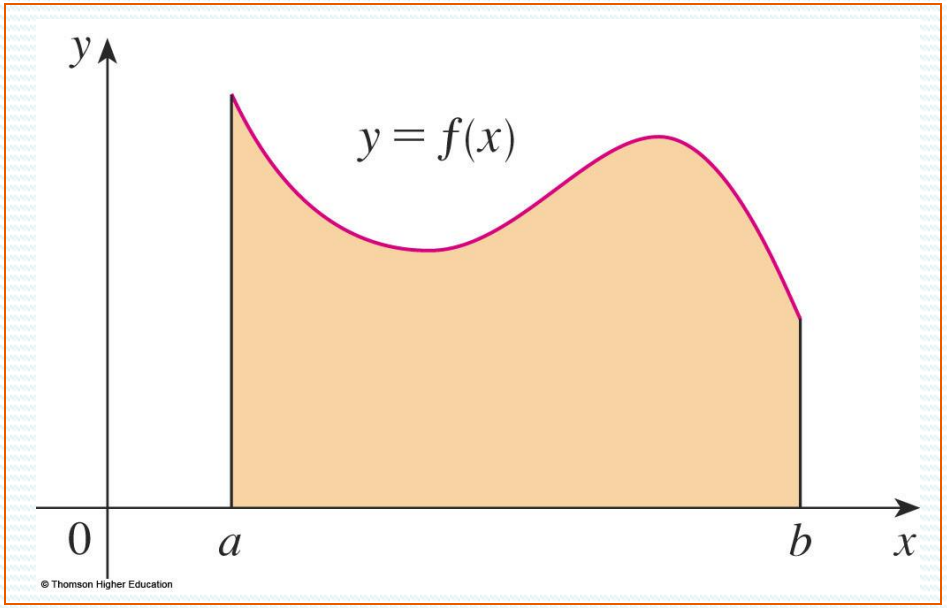
- We could use any letter in place of *x* without changing the value of the integral:

$$\int_a^b f(x)dx = \int_a^b f(t)dt = \int_a^b f(r)dr$$

Remark

Comparing above definition with the definition of area in Section 5.1, we see that the definite integral $\int_a^b f(x)\,dx$ can be interpreted as:

The area under the curve $y = f(x)$ from a to b

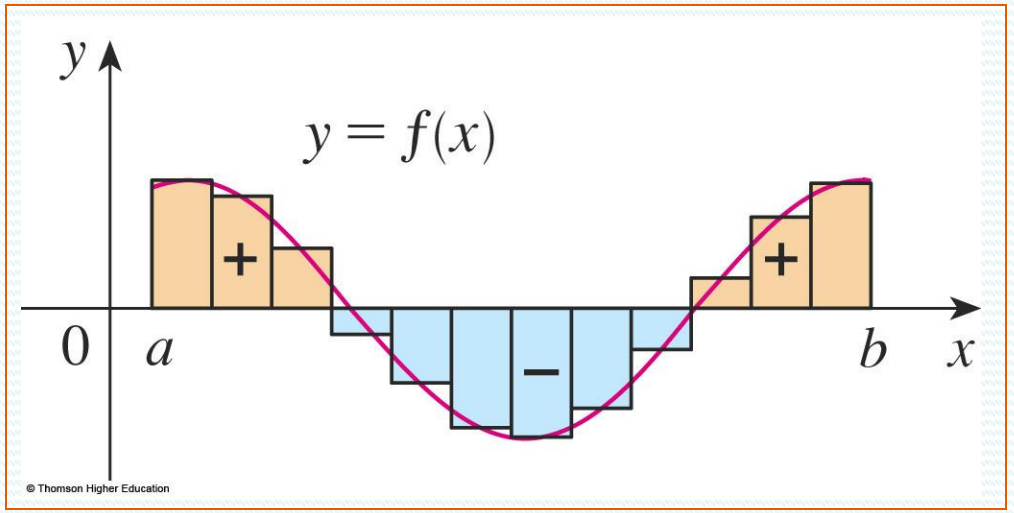


RIEMANN SUM

Note

If *f* takes on both positive and negative values, then the Riemann sum is:

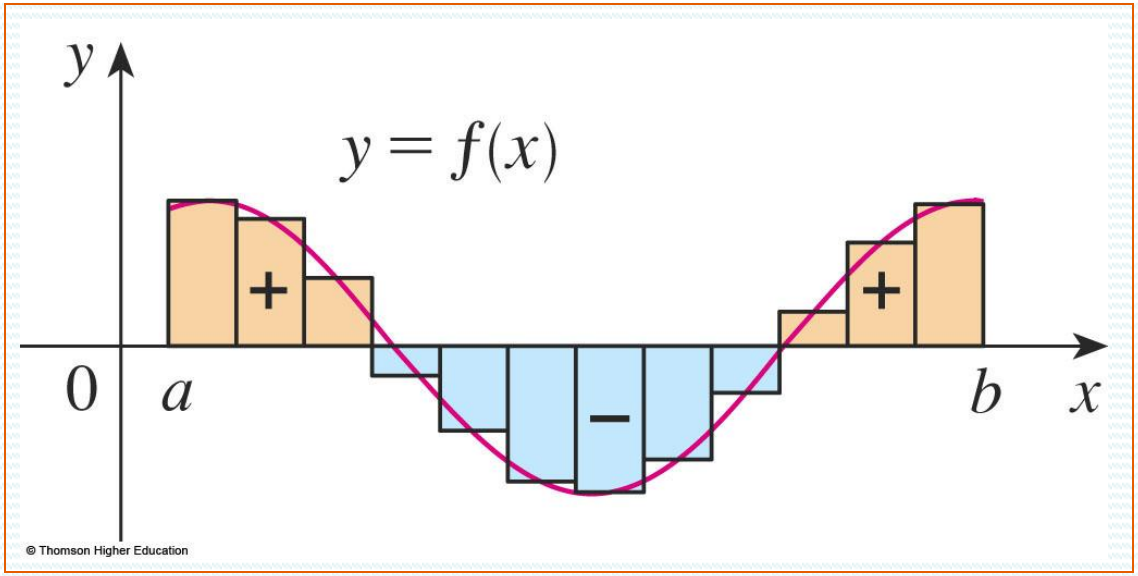
- The sum of the areas of the rectangles that lie above the *x*-axis and the negatives of the areas of the rectangles that lie below the *x*-axis
- That is, the areas of the gold rectangles minus the areas of the blue rectangles



RIEMANN SUM

Note

When we take the limit of such Riemann sums, we get the situation illustrated here.



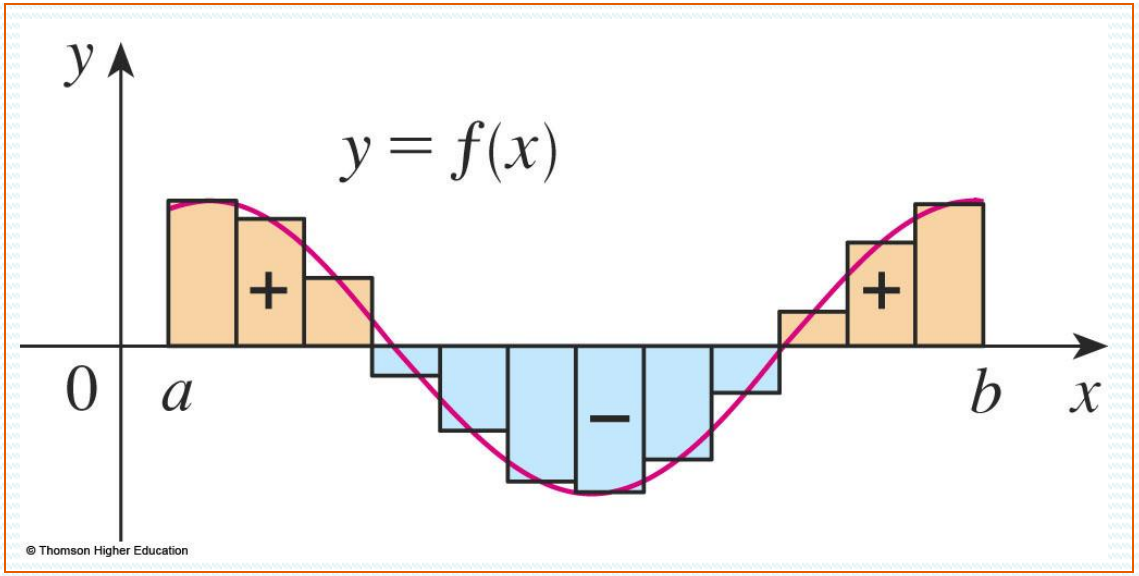
NET AREA

Note

A definite integral can be interpreted as a net area, that is, a difference of areas:

- A_1 is the area of the region above the x-axis and below the graph of f.
- A_2 is the area of the region below the x-axis and above the graph of f.

$$\int_a^b f(x)\,dx = A_1 - A_2$$



Theorem

If *f* is continuous on [*a*, *b*], or if *f* has only a finite number of jump discontinuities, then *f* is integrable on [*a*, *b*].

That is, the definite integral $\int_a^b f(x)\,dx$ exists.

Theorem

If *f* is integrable on [*a*, *b*], then

$$\int_a^b f(x)\,dx = \lim_{n_i \to \infty} \sum_{i=1}^{n} f(x_i)\,\Delta x$$

Where, $\Delta x = \dfrac{b-a}{n}$ and $x_i = a + i\,\Delta x$

EVALUATING INTEGRALS

$$\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$$

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\sum_{i=1}^{n} i^3 = \left[\frac{n(n+1)}{2}\right]^2$$

$$\lim_{n\to\infty} \frac{1}{n} = 0$$

Rules of sigma notation

$$\sum_{i=1}^{n} c = nc$$

$$\sum_{i=1}^{n} ca_i = c\sum_{i=1}^{n} a_i$$

$$\sum_{i=1}^{n} (a_i + b_i) = \sum_{i=1}^{n} a_i + \sum_{i=1}^{n} b_i$$

$$\sum_{i=1}^{n} (a_i - b_i) = \sum_{i=1}^{n} a_i - \sum_{i=1}^{n} b_i$$

Example

Find the area of the curve $f(x) = 5x - 2$

$a = 1$, $b = 4$.

Example

Find the area of the curve $f(x) = x^3$

$a = 0$, $b = 8$.

Example

Express $\lim_{n\to\infty} \sum_{i=1}^{n} (x_i^{\,3} + x_i \sin x_i)\Delta x_i$

as an integral on the interval [0, π].

Solution:

We are given that $a = 0$ and $b = \pi$.

So, by The theorem above, we have:

$$\lim_{n\to\infty} \sum_{i=1}^{n} (x_i^{\,3} + x_i \sin x_i)\,\Delta x_i = \int_0^{\pi} (x^3 + x \sin x)\,dx$$

PROPERTIES OF DEFINITE INTEGRAL

When we defined the definite integral $\int_a^b f(x)\,dx$, we implicitly assumed that $a < b$.

However, the definition as a limit of Riemann sums makes sense even if $a > b$.

PROPERTIES OF DEFINITE INTEGRAL

Notice that, if we reverse a and b, then Δx changes from $(b - a)/n$ to $(a - b)/n$.

Therefore, $\int_b^a f(x)\,dx = -\int_a^b f(x)\,dx$

If $a = b$, then $\Delta x = 0$, and so $\int_b^a f(x)\,dx = 0$

PROPERTIES OF INTEGRALS

We now develop some basic properties of integrals that will help us evaluate integrals in a simple manner.

PROPERTIES OF THE INTEGRAL

We assume *f* and *g* are continuous functions.

1. $\int_a^b c\,dx = c(b-a)$, where c is any constant

2. $\int_a^b [\,f(x)+g(x)]\,dx = \int_a^b f(x)\,dx + \int_a^b g(x)\,dx$

3. $\int_a^b c\,f(x)\,dx = c\int_a^b f(x)\,dx$, where c is any constant

4. $\int_a^b [\,f(x)-g(x)]\,dx = \int_a^b f(x)\,dx - \int_a^b g(x)\,dx$

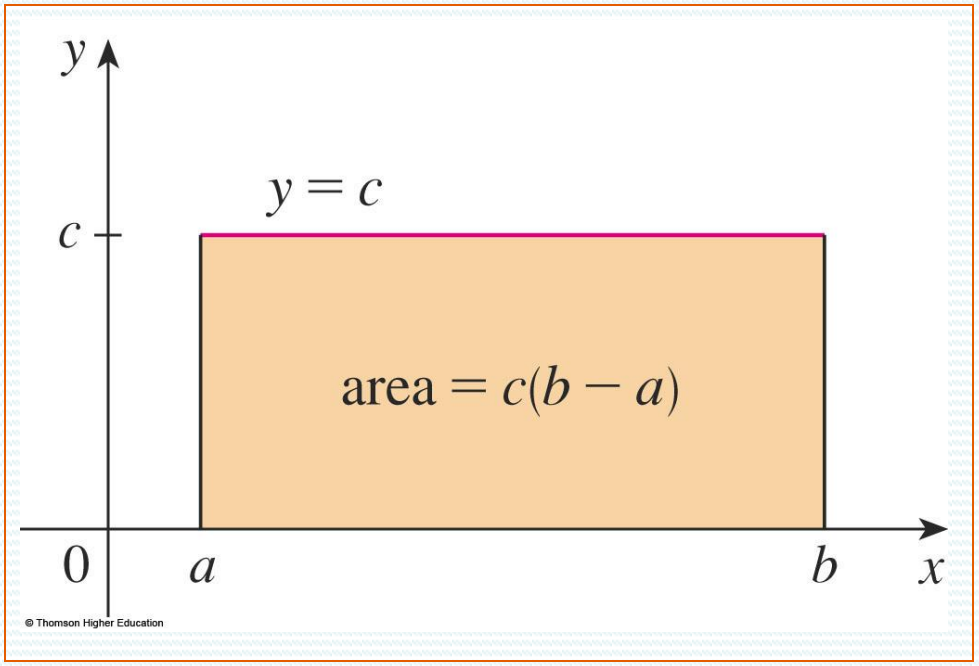
PROPERTY 1

$$\int_a^b c\,dx = c(b-a), \quad \text{where c is any constant}$$

Property 1 says that the integral of a constant function $f(x) = c$ is the constant times the length of the interval.

PROPERTY 1

If $c > 0$ and $a < b$, this is to be expected, because $c(b - a)$ is the area of the shaded rectangle here.



PROPERTY 2

$$\int_a^b [\, f(x) + g(x) \,] \, dx = \int_a^b f(x) \, dx + \int_a^b g(x) \, dx$$

Property 2 says that the integral of a sum is the sum of the integrals.

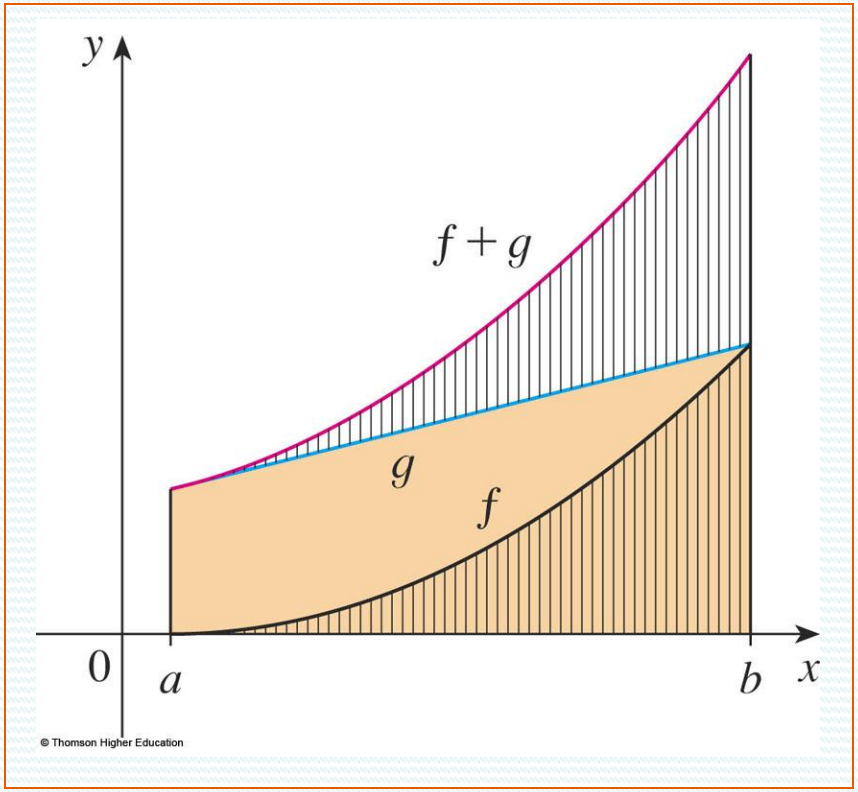
PROPERTY 2

For positive functions, it says that the area under $f + g$ is the area under f plus the area under g.

PROPERTY 2

The figure helps us understand why this is true.

- In view of how graphical addition works, the corresponding vertical line segments have equal height.



PROPERTY 3

$$\int_a^b c\, f(x)\, dx = c\int_a^b f(x)\, dx, \text{ where c is any constant}$$

Property 3 can be proved in a similar manner and says that the integral of a constant times a function is the constant times the integral of the function.

- That is, a constant (but only a constant) can be taken in front of an integral sign.

PROPERTY 4

$$\int_a^b [\, f(x) - g(x)]\, dx = \int_a^b f(x)\, dx - \int_a^b g(x)\, dx$$

Property 4 is proved by writing $f - g = f + (-g)$ and using Properties 2 and 3 with $c = -1$.

Example 6

Use the properties of integrals to evaluate $\int_0^1 (4+3x^2)\,dx$

- Using Properties 2 and 3 of integrals, we have:

$$\int_0^1 (4+3x^2)\,dx = \int_0^1 4\,dx + \int_0^1 3x^2\,dx$$

$$= \int_0^1 4\,dx + 3\int_0^1 x^2\,dx$$

PROPERTY 5

Property 5 tells us how to combine integrals of the same function over adjacent intervals:

$$\int_a^c f(x)\,dx + \int_c^b f(x)\,dx = \int_a^b f(x)\,dx$$

PROPERTY 5

In general, Property 5 is not easy to prove.

PROPERTIES OF INTEGRALS

Example 7

If it is known that

$$\int_0^{10} f(x)\,dx = 17 \quad \text{and} \quad \int_0^{8} f(x)\,dx = 12$$

find:

$$\int_8^{10} f(x)\,dx$$

Example 7

By Property 5, we have:

$$\int_0^8 f(x)\,dx + \int_8^{10} f(x)\,dx = \int_0^{10} f(x)\,dx$$

So,

$$\int_8^{10} f(x)\,dx = \int_0^{10} f(x)\,dx - \int_0^8 f(x)\,dx$$

$$= 17 - 12$$

$$= 5$$

COMPARISON PROPERTIES OF THE INTEGRAL

These properties, in which we compare sizes of functions and sizes of integrals, are true only if $a \le b$.

6. If $f(x) \ge 0$ for $a \le x \le b$, then $\int_a^b f(x)\,dx \ge 0$

7. If $f(x) \ge g(x)$ for $a \le x \le b$, then $\int_a^b f(x)\,dx \ge \int_a^b g(x)\,dx$

8. If $m \le f(x) \le M$ for $a \le x \le b$, then

$$m(b-a) \le \int_a^b f(x)\,dx \le M(b-a)$$

Example

Write as a single integral in the form